

M13-C3 — H4 cautious vs brave roots a/b on the closed H0 table

H4 run / analysed / not a Bucket 3 claim

1 Status

H4 run / analysed. Repository `baseline_cautious` vs locked ECAI brave on fixture `m13_bucket3_binary_collider_and / n30_seed42` is complete. Markdown twin: `h4_cautious_vs_brave.md`.

Lead finding: switching to cautious **blocks the brave residual-assumption gadget** that had allowed roots a/b to **solved**; paths coincide through early α_1 – α_3 setup, then brave says **OK** and finishes while cautious says **KO** and returns `completed_no_solution`. **Control:** child c still solves with a byte-identical delta to locked ECAI c . H0–H3 remain closed. **Not a Bucket 3 claim.** H4b analysed separately (`h4b_cautious_split_under_a.md`); H5 and H6 analysed separately (`h5_greedy_cautious.md`, `h6_folding_and_n_ablation.md`); H7a analysed separately (`h7a_relto_vs_sechk.md`); H7b not started.

2 Research question

On the closed H0 AND-collider table, does repository-baseline cautious learning change the ECAI-brave root outcomes for a/b that relied on a residual per-row assumption gadget (rows 9 vs 26), and what happens to control c ?

3 Main finding (lead)

Locked ECAI brave solves roots a/b with assumption-rich α -nests. Repository `baseline_cautious` on the **same** table returns `completed_no_solution` for both roots (no `delta.ab`).

Target	Locked ECAI brave	<code>baseline_cautious</code>
a	<code>solved</code> (α -nest)	<code>completed_no_solution</code> (~ 70 s; ~ 5 k-line stdout)
b	<code>solved</code> (mirror)	<code>completed_no_solution</code> (~ 66 s)
c (control)	<code>solved</code>	<code>solved</code> (byte-identical delta; ~ 1.3 s; audit SAT)

Learning-mode / entailment semantics change the root outcome by rejecting the 9-vs-26-style per-row assumption cover. This is **not** “cautious never solves” (see control c).

4 Decisive procedural fork

Both arms share the early path through α_1 – α_3 . Table conflict: row 9 is E^+ (1,0,0); row 26 is E^- (0,0,0) — identical BK `b_val_0,c_val_0`, opposite labels.

Brave:

```
folding result: c_alpha_3(A) <- [b_val_0(A)]
found: alpha_2/1 ... OK, assumption introduction result:
  c_alpha_3(A) <- [alpha_2(A),b_val_0(A)]
```

Closing nest:

```
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

Cautious:

```
folding result: c_alpha_3(A) <- [b_val_0(A)]
found: alpha_2/1 ... KO, cannot introduce an assumption for [b_val_0(A)]
```

then budget exhaustion and * No solution found!. Target b mirrors. Paths coincide until the brave gadget; semantics then diverge.

5 Control c

Cautious c still solves with a byte-identical delta to locked ECAI c :

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

Mode change is selective. String match / `solved` is not claimed as causal recovery.

6 Setup / paths

Fixture/sample: closed H0 `m13_bucket3_binary_collider_and` / `n30_seed42` (sha256 `1edae465...a8ca`). Brave arm: locked `ecai2024`. Cautious arm: `baseline_cautious` (`configs/baseline_cautious_config.pl`; not ECAI/AAMAS/H5). Primary targets a, b ; control c .

7 Semantic boundary

Authoritative for acceptance: Prolog learner outcome under the selected `learning_mode`. Under cautious learning, SAT of `bk.sol_chk.asp` is an existential final-artefact integrity witness, not proof that every positive is a cautious consequence.

8 Boundaries / non-claims

- Not a Bucket 3 claim.
- Cautious root no-solution is not mechanism recovery.
- Brave root **solved** is not mechanism recovery.
- Control *c* string match is not causal recovery.
- Not a ranking of brave vs cautious as generally better.
- H5, H6, and H7a analysed separately (`h5_greedy_cautious.md`, `h6_folding_and_n_ablation.md`, `h7a_relto_vs_sechk.md`); H7b not started. H6/H7a are not this H4 evidence body.

9 Prompted follow-ups (signpost only)

H5 is now run / analysed separately (`h5_greedy_cautious.md`). H6 (length drivers: *n*, `folding_steps` under fixed cautious) and H7a (brave nd `asm_intro(sechk)` vs `relto`) were prompted by H4 traces and are now run / analysed in separate records. H7b is not started.

10 Next

H4 documentation is complete. H4b and H5 are run / analysed separately. H6 and H7a are analysed separately; H7b is not started. Still **no Bucket 3 claim**.